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Insight

Connecting **2.5M+ rural beekeepers** directly to conscious consumers through **zero-modification hive sensing, 5-second optical purity verification**, and mass-conserved tamper-proof blockchain provenance.

Problem Statement ID – 26021

Team ID – 167577

Team Name – Arise

PS Category- Software

Theme - Agriculture, FoodTech & Rural Development

Problem Statement Title - Honey Chain: A block chain-based system for honey traceability and smart beekeeping management.



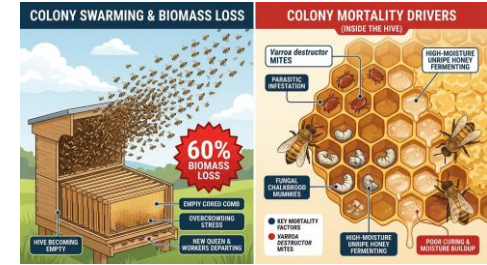
Ground Analysis & Core Breakthroughs:



1_honey_adulteration_problem



2_traditional_wall_hive_jalokha3

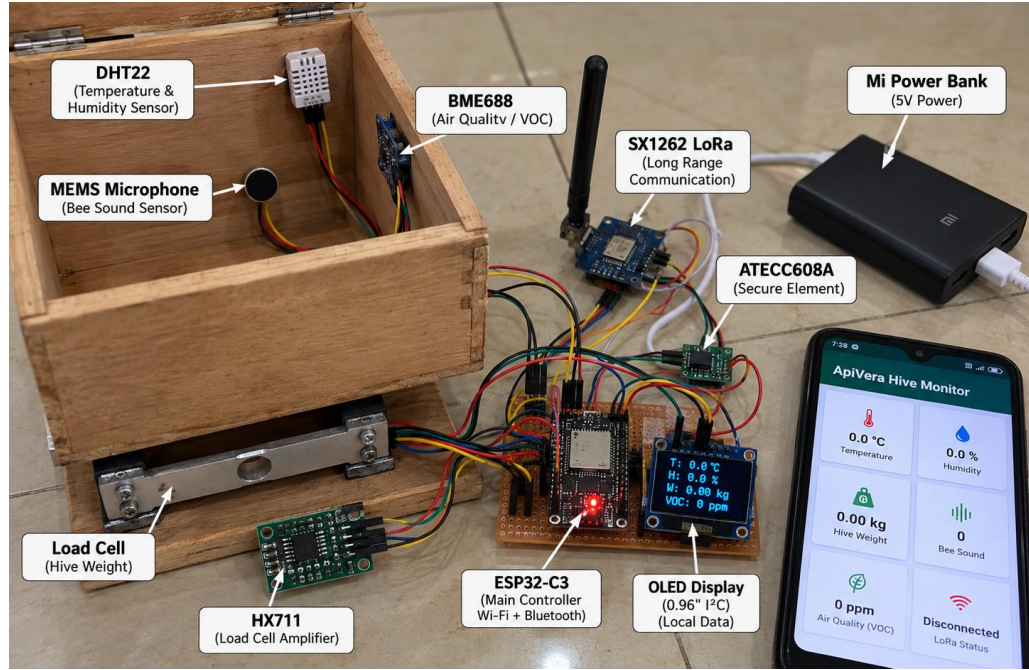


3_colony_swarming_mortality

- **Adulteration Defense:** Bypasses ₹20,000 lab tests using **5-Second Optical Polarimetry** ($[\alpha]_D$) to instantly detect C3/C4 & rice syrups.
- **Anti-Clone & Anti-Refill:** Enforces **Mass-Conservation Smart Contracts** (total jars = harvest scale weight) + **Burn-On-Open** single-use tokens.
- **100% Traditional Hive Fit: ApiVera Pod** retrofits onto Himalayan Wall Hives (*Jalokhas*), Log Hives, Mud Pots, and old boxes in 10 seconds with zero drilling.
- **Bio-Acoustic Pre-Emption:** Detects 240–280 Hz **queen piping** to warn farmers 48 hours before swarming via **regional WhatsApp voice notes**.

Introducing our Solution – APIVERA CHAIN

Indigenous Bio-Optical & Mass-Conserved Honey Provenance Ecosystem



Prototype V1: Testing



How it addresses the problem

- **Stops Adulteration:** Detects synthetic C3/C4 syrups in-field, eliminating reliance on ₹20,000 lab NMR tests.
- **Cuts Middlemen:** Direct farm-to-consumer traceability increases beekeeper earnings from ₹90/kg to ₹380/kg.
- **Prevents Colony Loss:** Delivers native-language voice alerts **48 hours before swarming** to save bee biomass.



Proposed Solution

Hive Layer: *ApiVera Pod* monitors acoustic frequencies, weight, and VOC gases to predict swarms and brood diseases.

Harvest Layer: *ApiVera OptiLens* verifies raw honey purity in 5 seconds via a ₹50 smartphone optical clip.

Retail Layer: *ApiVera Ledger* locks minted jars to harvested weight and burns tokens on first open to stop bottle refilling.



Innovation & Uniqueness



5-Second Optical Purity Check: Uses light rotation on a ₹50 mobile lens to instantly catch fake sugar syrups without expensive lab tests.



Anti-Refill Smart Seal: Caps total jar QR codes to actual harvest weight and burns the digital token on first open to stop bottle refilling.



Bio-Acoustic Swarm Warning: Listens to bee sound frequencies to alert the farmer **48 hours before bees escape** (swarming).

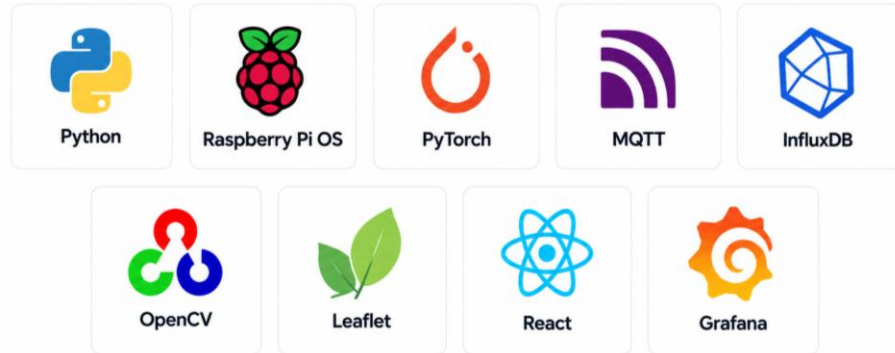


Smart Moisture Tracker: Measures nightly water evaporation to notify the farmer when honey is **100% ripe ($\leq 18\%$ moisture)** before harvest.



Zero-App Voice & Direct UPI Pay: Sends automated audio alerts in regional languages (Hindi, Punjabi, etc.) and allows consumers to **tip the beekeeper directly via UPI**.

Technological Stack



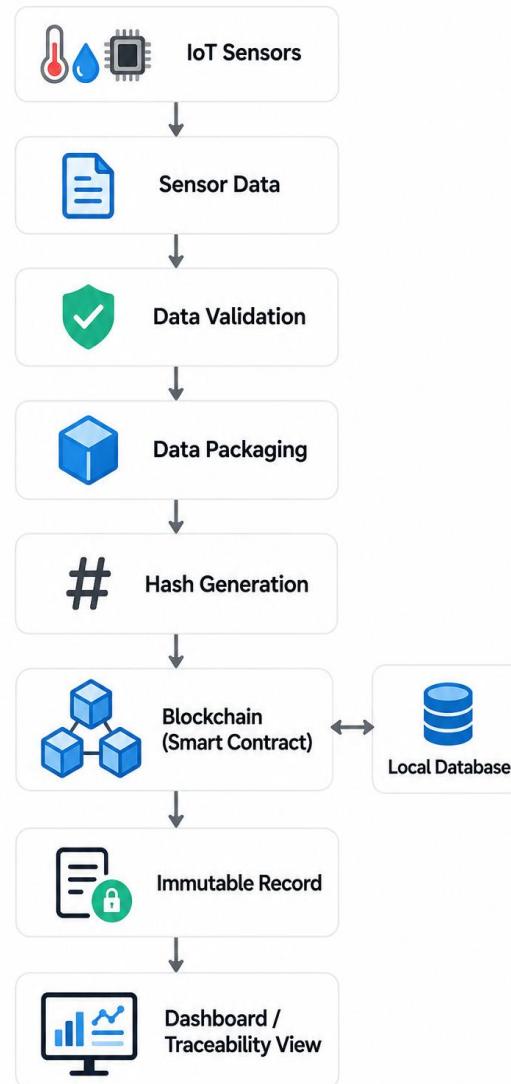
Hardware Stack (Offline Edge Intelligence)



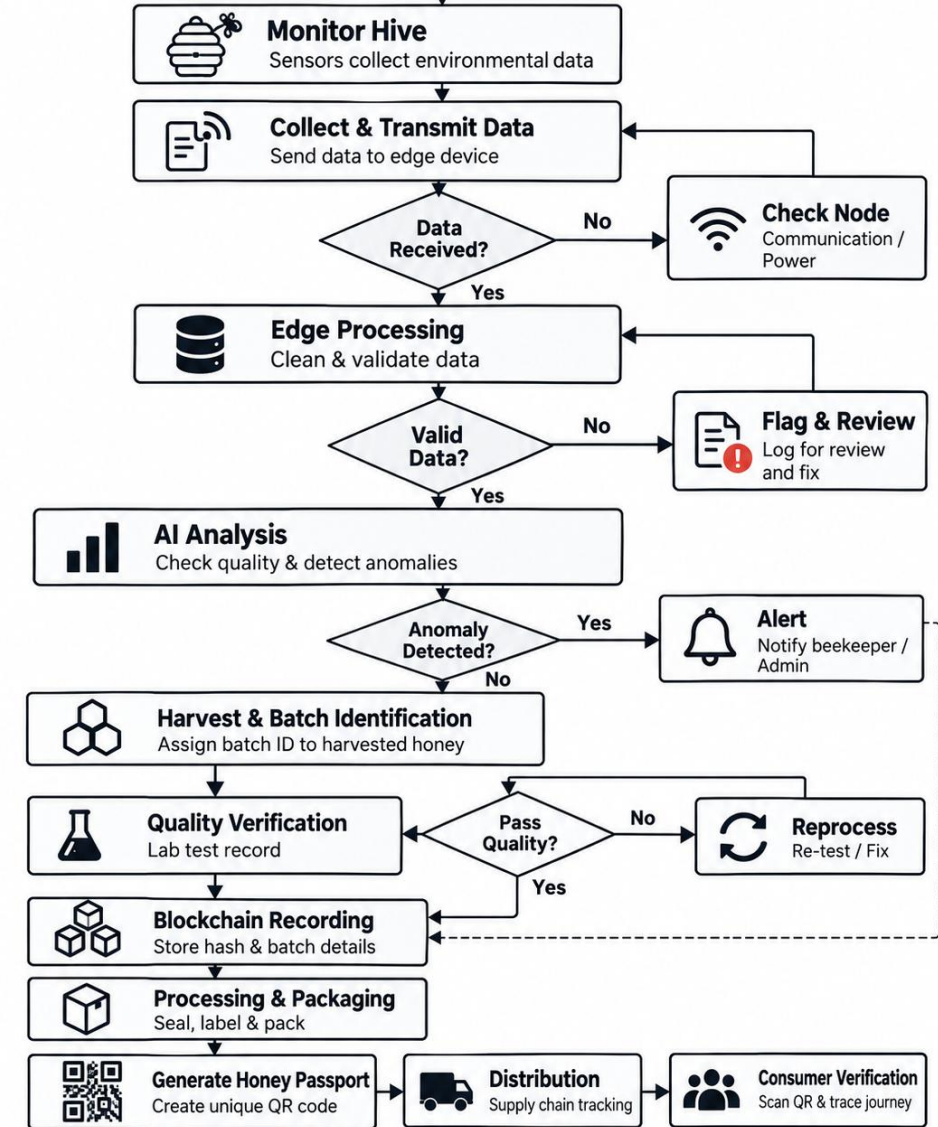
Blockchain Stack



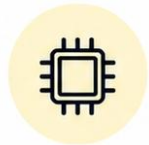
Blockchain Integration



End-to-End Honey Traceability Flow & Methodology



Deployment & Scale Viability



Prototype Ready

Uses off-the-shelf hardware for rapid field validation.



Cluster Architecture

One gateway can serve multiple nearby hives.



Modular Expansion

Start with essential sensors and add modules as needed.



Retrofit Deployment

Works with existing hives without major changes.



Offline-First Operation

Stores data locally and syncs when network is available.



Scalable Deployment

From single hive to cooperatives and national reach.

Deployment Roadmap



Prototype
1 Hive



Pilot Cluster
10 Hives



FPO Integration
Cooperative



Processor Integration
Quality & Packaging



Scaled Deployment
Regional / National

Feasibility & Viability (Unique Operational Strengths)

- **Zero-New Infrastructure:** Plugs directly into existing **KVIC Honey Mission extraction toolkits** and Khadi village centers.
- **Micro-Gas Blockchain Batches:** Bundles 100+ hive harvests into a single Merkle cryptographic root, costing **under ₹0.05 per batch**.
- **Self-Calibrating Bio-Sensing:** Automatically filters out ambient tractor, wind, and rain noise without manual tuning.
- **Migratory Auto-Tracking:** Automatically updates floral origin and GPS when beekeepers move boxes between states (e.g., Punjab Mustard → Bihar Litchi).

Unique Ground Challenges & Technical Mitigations

- **State Migrations on Trucks → Solved** via **autonomous GPS geofencing** that auto-switches botanical flora profiles during seasonal transit.
- **Monsoon Moisture Spoilage (> 20%) → Solved** via **smart contract lock** that physically refuses to mint QR tokens if moisture exceeds 18.0%.
- **Extreme Seasonal Temperature Drift → Solved** via **polynomial thermal compensation** keeping scale accuracy within $\pm 2.5g$ from $-20^{\circ}C$ to $+48^{\circ}C$.
- **Fake Lab Certificate Uploads → Solved** via **hardware secure element (ATECC608A)** signing test data directly at the village refractometer.



1. Hive Monitoring



Temp Humidity Weight Acoustic

2. Harvest Recording



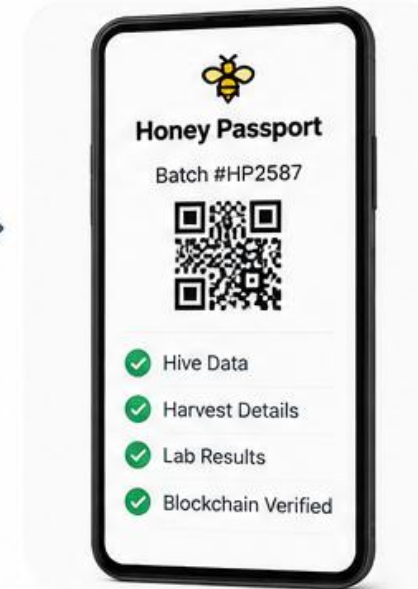
Weight Flow Quality

3. Quality Testing

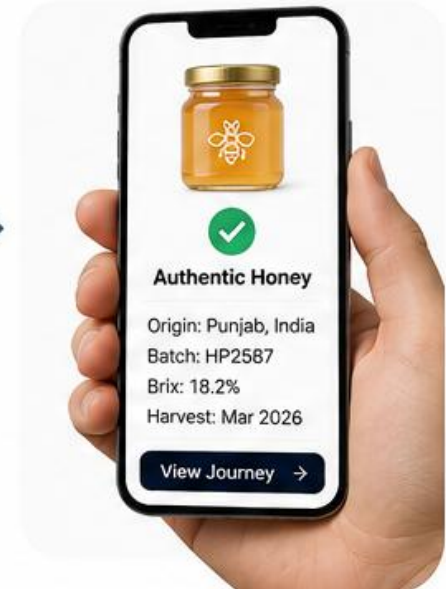


Brix pH Authenticated

4. Blockchain Record



5. Consumer Verification



For Beekeepers

Higher productivity and income

↑
+30%

Hive
productivity

↑
+20%

Income
per harvest

↑
-50%

Hive
losses



For Consumers

Trusted and safe honey

↑
100%

Traceable origin
(QR Honey Passport)

↑
100%

Quality verified
batches



Economic Impact

Stronger rural economy

↑
+25%

Higher market
value for
verified honey

↑
-60%

Manual record
keeping effort

↑
More

Market access
for FPOs &
cooperatives



Environmental Impact

Sustainable and resilient
ecosystems

↑
-40%

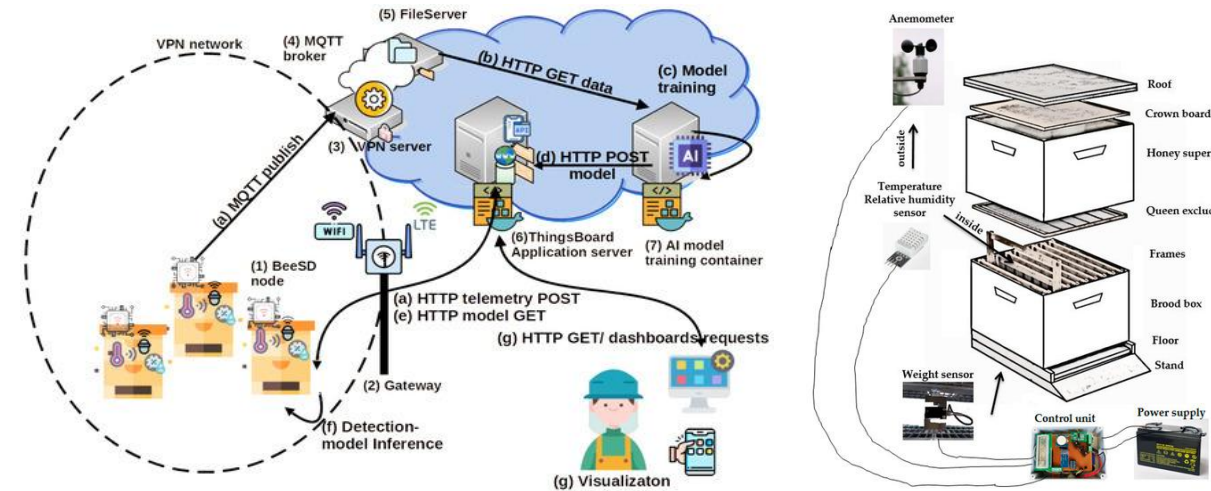
Unnecessary
hive interventions

↑
Better

Pollinator health
and biodiversity

Comprehensive Scientific References

#	Research / Reference	What it supports	Link
01	Honey Traceability & Authenticity Review - Journal of Apicultural Science (2022)	Honey origin authentication, IoT & blockchain applications	Research Paper
02	Fujairah Honey Chain - Information (2025)	Blockchain + IoT + smart contracts + honey batch traceability	Research Paper
03	Honey Authentication using AI-Based Pollen Analysis - British Food Journal (2025)	AI / ML for honey authentication and pollen analysis	Research Paper
04	Cutting-edge Approaches for Honey Authentication - JFCA (2025)	Chemical, molecular & AI-based authentication	Research Paper
05	Precision Apiculture System - Sensors (2020)	Temperature, humidity, weight & hive monitoring using IoT	Research Paper / System
06	KVIC Honey Mission - Government of India	Existing beekeeping ecosystem, training, bee boxes & toolkits	KVIC Honey Mission Document



A. Smart Detection system's high-level architecture. Bee Smart Detection system parts, data inputs, outputs, and data processing steps

B. prototypes tested in the beekeeping station of the laboratory team of MicroComputer Systems, Department of Mathematics, University of Ioannina, located in the Ligopsa area, Ioannina, Epirus, Greece.

